

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Isobutyric anhydride**  
 Product Description: **Isobutyric anhydride**  
 Cat No. : 96251  
 Synonyms 2-Methylpropionic anhydride  
 CAS No 97-72-3  
 Molecular Formula C<sub>8</sub> H<sub>14</sub> O<sub>3</sub>

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company** Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

**E-mail address** Enquiry.my@thermofisher.com

**Emergency Telephone Number** Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                    |                     |
|------------------------------------|---------------------|
| Acute dermal toxicity              | Category 3 (H311)   |
| Acute Inhalation Toxicity - Vapors | Category 3 (H331)   |
| Skin Corrosion/Irritation          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation  | Category 1 (H318)   |

**Label Elements**


# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

## Signal Word

## Danger

### Hazard Statements

H314 - Causes severe skin burns and eye damage  
H311 + H331 - Toxic in contact with skin or if inhaled

### Precautionary Statements

#### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P374 - Fight fire with normal precautions from a reasonable distance  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P235 - Store in a well-ventilated place. Keep cool  
P404 - Store in a closed container  
P405 - Store locked up

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Combustible liquid

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component            | CAS No  | Weight % |
|----------------------|---------|----------|
| Isobutyric anhydride | 97-72-3 | >95      |
| Propionic acid       | 79-09-4 | <=1      |
| Acetic acid          | 64-19-7 | <=0.5    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

**Inhalation** Remove to fresh air. If breathing is difficult, give oxygen. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

Difficulty in breathing. Causes burns by all exposure routes. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Water.

**Special hazards arising from the substance or mixture**

Combustible material. Flammable. Vapors may form explosive mixtures with air. Containers may explode when heated.

**Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Avoid contact with skin, eyes or clothing. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental precautions**

See Section 12 for additional Ecological Information.

**Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Do not let this chemical enter the environment. Remove all sources of ignition.

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep away from open flames, hot surfaces and sources of ignition. Avoid contact with skin, eyes or clothing. Avoid breathing dust/fume/gas/mist/vapors/spray. Avoid ingestion and inhalation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep away from heat, sparks and flame. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component      | Malaysia | ACGIH TLV                   | OSHA PEL                                                                                                 |
|----------------|----------|-----------------------------|----------------------------------------------------------------------------------------------------------|
| Propionic acid |          | TWA: 10 ppm                 | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 30 mg/m <sup>3</sup>                                             |
| Acetic acid    |          | TWA: 10 ppm<br>STEL: 15 ppm | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 25 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> |

| Component      | European Union                                                                                                   | The United Kingdom                                                                                             | Germany                                                                                                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Propionic acid | TWA: 10 ppm (8h)<br>TWA: 31 mg/m <sup>3</sup> (8h)<br>STEL: 20 ppm (15min)<br>STEL: 62 mg/m <sup>3</sup> (15min) | STEL: 15 ppm 15 min<br>STEL: 46 mg/m <sup>3</sup> 15 min<br>TWA: 10 ppm 8 hr<br>TWA: 31 mg/m <sup>3</sup> 8 hr | TWA: 10 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 31 mg/m <sup>3</sup> (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK<br>TWA: 31 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 62 mg/m <sup>3</sup> |
| Acetic acid    | TWA: 25 mg/m <sup>3</sup> (8h)<br>TWA: 10 ppm (8h)<br>STEL: 50 mg/m <sup>3</sup> (15min)<br>STEL: 20 ppm (15min) | STEL: 37 mg/m <sup>3</sup><br>STEL: 15 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup>                         | TWA: 10 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 25 mg/m <sup>3</sup> (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK<br>TWA: 25 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 50 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

Eye Protection  
Hand Protection

Goggles  
Protective gloves

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

|                                 |                                                                          |
|---------------------------------|--------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure |
|---------------------------------|--------------------------------------------------------------------------|

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>   | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                                                          |
| <b>Recommended Filter type:</b> | Organic gases and vapours filter Type A Brown conforming to EN14387<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |

|                         |                                                                       |
|-------------------------|-----------------------------------------------------------------------|
| <b>Hygiene Measures</b> | Handle in accordance with good industrial hygiene and safety practice |
|-------------------------|-----------------------------------------------------------------------|

|                                        |                          |
|----------------------------------------|--------------------------|
| <b>Environmental exposure controls</b> | No information available |
|----------------------------------------|--------------------------|

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                |                                                 |                                          |
|------------------------------------------------|-------------------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                                       |                                          |
| <b>Physical State</b>                          | Liquid                                          |                                          |
| <b>Odor</b>                                    | pungent                                         |                                          |
| <b>Odor Threshold</b>                          | No data available                               |                                          |
| <b>pH</b>                                      | No information available                        |                                          |
| <b>Melting Point/Range</b>                     | -53 °C / -63.4 °F                               |                                          |
| <b>Softening Point</b>                         | No data available                               |                                          |
| <b>Boiling Point/Range</b>                     | 182 °C / 359.6 °F                               | @ 760 mmHg                               |
| <b>Flash Point</b>                             | 66.4 °C / 151.5 °F                              | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available                               |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                                  | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.09 Vol%<br><b>Upper</b> 7.7 Vol% |                                          |
| <b>Vapor Pressure</b>                          | 0.7 mbar @ 20 °C                                |                                          |
| <b>Vapor Density</b>                           | No information available                        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 0.954                                           |                                          |
| <b>Bulk Density</b>                            | Not applicable                                  | Liquid                                   |
| <b>Water Solubility</b>                        | Decomposes                                      |                                          |
| <b>Solubility in other solvents</b>            | No information available                        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                  |                                          |
| Isobutyric anhydride                           | 1.1                                             |                                          |
| Propionic acid                                 | 0.25 - 0.33                                     |                                          |
| Acetic acid                                    | -0.2                                            |                                          |

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

Autoignition Temperature 329 °C / 624.2 °F  
Decomposition Temperature No data available  
Viscosity 1.18 cP (25°C)  
Explosive Properties explosive air/vapour mixtures possible  
Oxidizing Properties No information available

Molecular Formula C8 H14 O3  
Molecular Weight 158.2

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Moisture sensitive.

### Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous polymerization does not occur.  
Hazardous Reactions No information available.

### Conditions to Avoid

Heat, flames and sparks. Incompatible products. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

Strong oxidizing agents. Strong bases. Alcohols.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

Product Information No acute toxicity information is available for this product

#### (a) acute toxicity;

Oral Based on available data, the classification criteria are not met  
Dermal Category 3  
Inhalation Category 3

| Component            | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                |
|----------------------|---------------------------|------------------------------|--------------------------------|
| Isobutyric anhydride | 2230 mg/kg (Rat)          | 474 mg/kg (Rabbit)           | 4.21 mg/L/7h (Rat)             |
| Propionic acid       | LD50 = 3455 mg/kg ( Rat ) | LD50 = 3235 mg/kg ( Rabbit ) | LC50 = > 19.7 mg/l ( Rat ) 1 h |
| Acetic acid          | 3310 mg/kg ( Rat )        | -                            | > 40 mg/L ( Rat ) 4 h          |

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

(f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Based on available data, the classification criteria are not met

(i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Other Adverse Effects

The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Do not empty into drains. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.

| Component            | Freshwater Fish                                                                                                                                                 | Water Flea           | Freshwater Algae                                                                                   | Microtox              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------|-----------------------|
| Isobutyric anhydride | Leuciscus idus: LC50= 146 mg/L/96h                                                                                                                              | EC50= 51.25 mg/L/48h |                                                                                                    |                       |
| Propionic acid       | LC50: = 51 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 73 - 99.7 mg/L, 96h static (Lepomis macrochirus)<br>LC50: > 1 mg/L, 96h static (Pimephales promelas) |                      | EC50: = 45.8 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: = 43 mg/L, 96h (Desmodesmus subspicatus) | EC50 = 59.6 mg/L 17 h |
| Acetic acid          | Pimephales promelas:                                                                                                                                            | EC50 = 95 mg/L/24h   | -                                                                                                  | Photobacterium        |

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

|  |                                                                  |  |  |                                                                                                                                                               |
|--|------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | LC50 = 88 mg/L/96h<br>Lepomis macrochirus:<br>LC50 = 75 mg/L/96h |  |  | phosphoreum: EC50 =<br>8.8 mg/L/15 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/25 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/5 min |
|--|------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Persistence and degradability

### **Persistence**

Persistence is unlikely, based on information available.

### **Degradability**

Decomposes in contact with water.

### **Degradation in sewage treatment plant**

No information available. Decomposes in contact with water.

## Bioaccumulative potential

Product does not bioaccumulate due to reaction with water

| Component            | log Pow     | Bioconcentration factor (BCF) |
|----------------------|-------------|-------------------------------|
| Isobutyric anhydride | 1.1         | No data available             |
| Propionic acid       | 0.25 - 0.33 | No data available             |
| Acetic acid          | -0.2        | No data available             |

## Mobility in soil

Decomposes in contact with water. . Is not likely mobile in the environment.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

### **Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

### **Other Information**

Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer Large amounts will affect pH and harm aquatic organisms

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

|                         |                                                        |
|-------------------------|--------------------------------------------------------|
| UN-No                   | UN2922                                                 |
| Hazard Class            | 8                                                      |
| Subsidiary Hazard Class | 6.1                                                    |
| Packing Group           | II                                                     |
| Proper Shipping Name    | Corrosive liquid, toxic, n.o.s. (ISOBUTYRIC ANHYDRIDE) |

### Road and Rail Transport

|                         |        |
|-------------------------|--------|
| UN-No                   | UN2922 |
| Hazard Class            | 8      |
| Subsidiary Hazard Class | 6.1    |
| Packing Group           | II     |

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

**Proper Shipping Name** Corrosive liquid, toxic, n.o.s. (ISOBUTYRIC ANHYDRIDE)

## IATA

**UN-No** UN2922  
**Hazard Class** 8  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II  
**Proper Shipping Name** Corrosive liquid, toxic, n.o.s. (ISOBUTYRIC ANHYDRIDE)

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component            | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Isobutyric anhydride | 202-603-6 | X    | X   | X     | X    | X    | X     | X    | KE-24876 |
| Propionic acid       | 201-176-3 | X    | X   | X     | X    | X    | X     | X    | KE-29352 |
| Acetic acid          | 200-580-7 | X    | X   | X     | X    | X    | X     | X    | X        |

| Component      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Propionic acid |                                                                                           |                                                                                          |                            | Annex I - Y34                      |
| Acetic acid    |                                                                                           |                                                                                          |                            | Annex I - Y34                      |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

# SAFETY DATA SHEET

Isobutyric anhydride

Revision Date 26-Mar-2025

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Prepared By

Revision Date

Revision Summary

Health, Safety and Environmental Department

26-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**